

① Solution of Exercise: 04 (Page: 5)

The largest profit = 69

The smallest profit = 16

Range = $69 - 16 = 53$

We, assume that, class number is = 5

So, class intervals is $= 53/5 = 10.6$, it's equivalent to 11.

frequency distribution

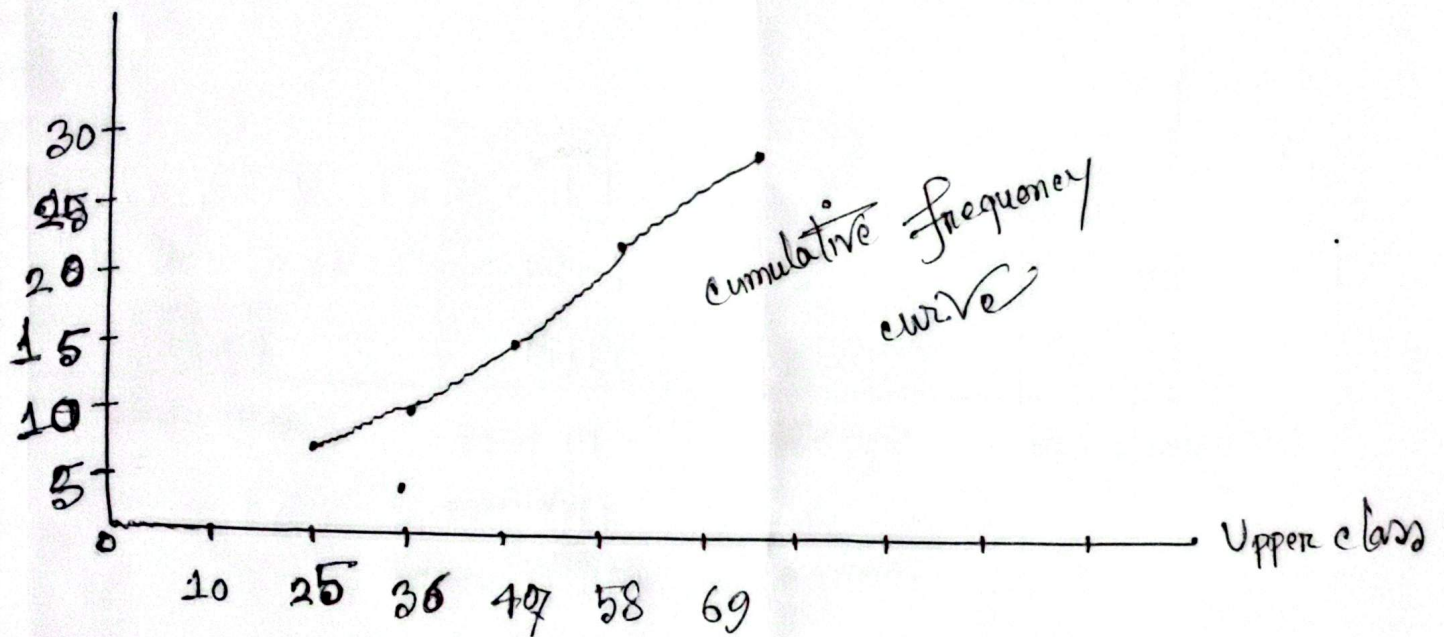
Class interval	Tally	Frequency
15 - 25		6
26 - 36		3
37 - 47		6
48 - 58		8
59 - 69		7

② Cumulative frequency curve: -

(ii) NO

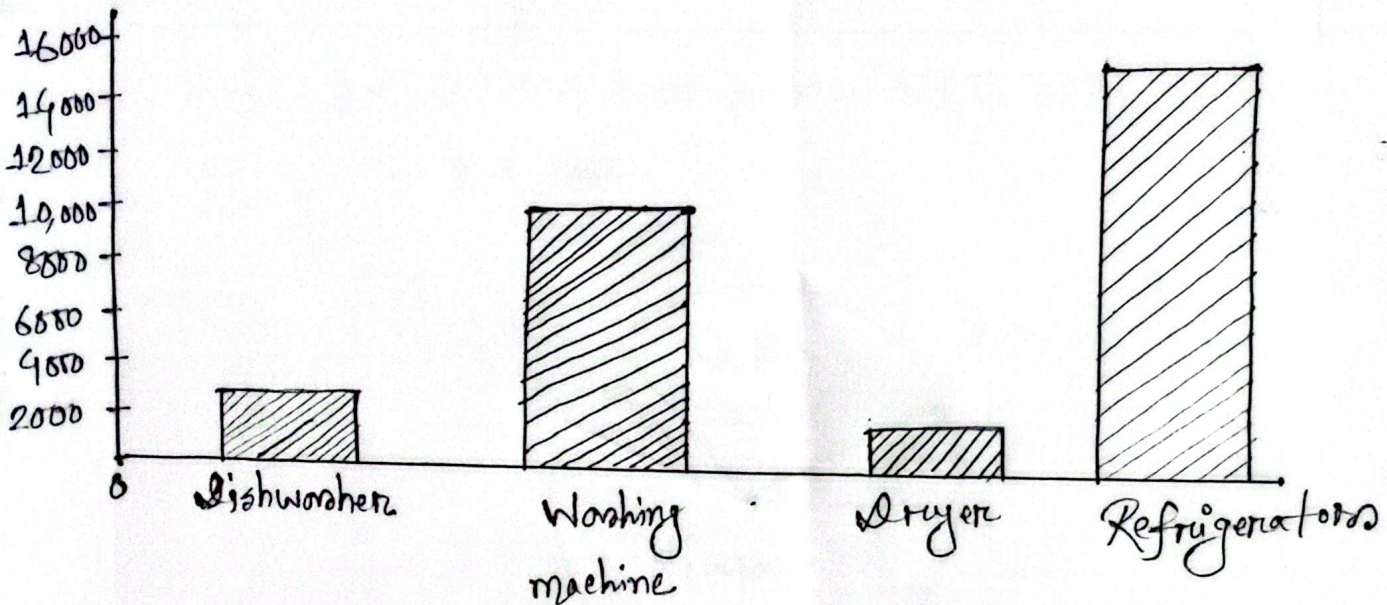
Cumulative frequency curve:-

class intervals	15-25	26-36	37-47	48-58	59-69
frequency	6	3	6	8	7
cumulative frequency	6	9	15	23	30



Exercise: 06 (Page: 10)

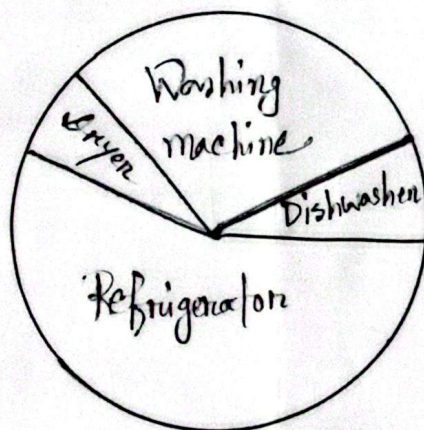
Products	Dishwasher	Washing machine	Dryer	Refrigerators
Price (TK)	2500	9000	2000	15500



Picture: Bar Diagram.

Products	Dishwasher	Washing machine	Dryer	Refrigerators
Price (₹)	2500	9000	2000	15,500
Amount of angle = $(f/n) \times 360$	31	112	25	192
Cumulative amount of angle	31	143	168	360

$$\begin{aligned} \text{Total, } n &= 2500 + 9000 + 2000 + 15,500 \\ &= 29,000 \end{aligned}$$



Picture: Pie Diagram.

Age	No. of worker (F)	Midvalue (x)	fx
18 - 22	120	20	2400
22 - 26	125	24	3000
26 - 30	280	28	7840
30 - 34	260	32	8320
34 - 38	155	36	5580
38 - 42	184	40	7360
42 - 46	162	44	7128
46 - 50	86	48	4128
50 - 54	75	52	3900
54 - 58	53	56	2968
n = 1500			

$$\text{Mean, } \bar{x} = \frac{\sum fx}{n}$$

$$= \frac{52,624}{1500}$$

$$= 35.08 \text{ years.}$$

* It's direct method of finding out mean.

4 नए कारवाहन
5 नए प्रयोग

अर्थात् निर्णय (Median)

Fuel	NO. of Car (f)	Cumulative frequency (cf)
0 - 100	50	50
100 - 200	125	175 (cf)
<u>200 - 300</u>	200 Fm	375
300 - 400	75	450
400 - 500	40	490
500 - 600	10	500
	n = 500	

$$\begin{aligned}
 \text{So, Median} &= L + \frac{\frac{n}{2} - cf}{f_m} \times c \\
 &= 200 + \frac{\frac{500}{2} - 175}{200} \times 100 \\
 &= 200 + \frac{250 - 175}{200} \times 100 \\
 &= 200 + 37.5 \times 100 \\
 &= 200 + 37.5 \\
 &= 237.5 \quad (\text{Ans})
 \end{aligned}$$

Here,

L = Lower limit of median class

n = Total number of observation.

Fm = Frequency of median class,

cf = Preceding cumulative frequency of median class.

c = class interval,